

Adnan Ahmed Khan

AI & Backend Engineer | Production LLM Systems, Agents & Cloud Platforms

Islamabad, Pakistan | khanadnanahmed01@gmail.com | [linkedin.com/in/adnahmed](https://www.linkedin.com/in/adnahmed) | github.com/adnahmed

PROFESSIONAL SUMMARY

AI & Backend Engineer with 6+ years building and operating Python/TypeScript services, distributed systems, and AWS platforms. Hands-on with LangGraph/LangChain agent workflows, OpenAI-compatible APIs, tool calling, structured outputs, model routing, embeddings/vector retrieval, and production-oriented AI automation. Scaled systems to 500-800 RPS and 50k-100k DAU; reduced query latency from 30s+ to under 1s, failed operations by 80%, AWS spend by 25%, and MTTR from about 45 minutes to under 10.

TECHNICAL SKILLS

AI / LLM: LLM applications, agentic and multi-agent workflows, RAG/vector retrieval, embeddings, prompt/context engineering, tool calling, structured outputs, OpenAI-compatible APIs, Anthropic model routing, evaluation/testing, LangGraph, LangChain, ChromaDB, Graphiti/Neo4j
Backend / APIs: Python 3.12, FastAPI, TypeScript, JavaScript, Node.js, Nest.js, Express, REST, async HTTP/httpx, microservices, workflow orchestration
Cloud / Data / Reliability: AWS EKS/ECS/Lambda/RDS, Docker, Kubernetes, Terraform, Helm, GitHub Actions, ArgoCD, PostgreSQL, Redis, Kafka, Elasticsearch, MongoDB, DynamoDB, idempotency, retries, distributed state, Prometheus, Grafana, Jaeger, pytest, Jest

SELECTED AI & LLM ENGINEERING

Verifiable Browser Agent Runtime | Python, LangGraph, FastAPI, Playwright, Chroma, Graphiti/Neo4j | [GitHub](#)

- Engineered an asynchronous agent platform with LangGraph control loops, grounded Playwright/CDP actions, pre/post semantic effect verification, bounded recovery, and multi-agent critic/reducer reasoning.
- Built an OpenAI-style FastAPI responses gateway with SSE streaming, tool calls, model/capability and quota routing, retries, transcripts, embeddings, and optional Chroma vector retrieval.
- Designed distributed workers with leases, heartbeats, encrypted shared state, replay/snapshots, GitHub Actions supervision, Docker delivery, and automated quality checks.

openclaw-workflow - Declarative Multi-Agent Orchestrator | TypeScript, Redis, MCP, JSON Schema | [GitHub](#)

- Architected YAML/JSON-driven LLM pipelines with dependency graphs, parallel execution, per-step model overrides, retries/timeouts, cancellation, output gates, cache adoption, and partial resume.
- Implemented runtime/legacy/CLI session adapters, filesystem/native Redis/MCP Redis state, semantic contracts, artifact-backed outputs, schema validation, and automated workflow tests.

PROFESSIONAL EXPERIENCE

HackOnTech | Senior Software Engineer, Backend & Platform | Remote / Islamabad | 2023 - Present

- Own core backend/platform engineering for B2B/G2G NGO systems serving 50k-100k daily users and 500-800 RPS peak traffic.
- Led a 0-to-1 overhaul into six high-throughput microservices, enabling zero-downtime releases while protecting payments, queues, and customer-facing availability.
- Engineered idempotent distributed workflows with retries, rollbacks, and recovery, cutting failed cross-service operations by 80% and payment failures from about 5% to 2%.
- Reduced PostgreSQL/Redis query latency from 30+ seconds to under 1 second through query redesign, indexing, and caching.
- Reduced AWS spend by 25% across EKS/autoscaling workloads and cut MTTR from about 45 minutes to under 10 with tracing, dashboards, and service-level observability.

Environment: Python, FastAPI, TypeScript, Node.js, Nest.js, AWS EKS/Lambda, PostgreSQL, Redis, Temporal, Step Functions, Terraform, Helm, ArgoCD, Prometheus, Grafana, Jaeger

Creative IT Park | Backend Developer | 2021 - 2023

- Delivered Python/FastAPI and Node.js platforms for government, university, and healthcare clients, owning production APIs and integrations.
- Hardened Kafka notification pipelines during downstream failures; optimized Elasticsearch/data access and caching to reduce public-portal latency.
- Secured healthcare ETL workflows with encrypted data flows and fine-grained AWS IAM controls for regulated and sensitive-data environments.

GeekGenix | Backend Developer | 2020 - 2021 - Built Express/MongoDB APIs, Stripe checkout and webhook reconciliation, and Passport.js authentication.

EDUCATION

BS Computer Science - PMAS Arid Agriculture University, Rawalpindi